



NIRMALA NCC AUTOMATION

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INTRODUCTION

1.1 Objective of the Project

The main objective of the Nirmala NCC project is to create a digital platform that streamlines the management of cadet activities, registrations, attendance, and uniform distribution within the NCC unit of Nirmala College. The project aims to eliminate manual paperwork and provide both cadets and administrators with a user-friendly system where all necessary information can be recorded, accessed, and updated easily. By implementing this project, the unit will be able to maintain a centralized database that ensures accuracy, transparency, and efficiency in handling all NCC-related operations.

The current system largely relies on manual records such as registers, paper applications, and offline communication, which often lead to delays, duplication of data, and difficulty in maintaining large volumes of information. Attendance is usually marked manually, bills are written on paper, and cadet details are scattered across different files. Although this method is simple and does not require technical expertise, it is time-consuming, prone to human error, and lacks proper data security and accessibility. As the number of cadets grows, these challenges make it harder for the staff to manage daily operations effectively.

The proposed Nirmala NCC management system overcomes these limitations by introducing a modern, web-based platform with features like secure registration, cadet approval by admin, automated attendance management, digital billing for uniforms, and dashboards for both cadets and administrators. This system offers several advantages, including faster processing, improved accuracy, data security, and anytime accessibility. It also provides a more professional and transparent way to manage cadet information, generate reports, and communicate updates. By shifting from manual to digital, the project not only saves time and resources but also enhances the overall efficiency and credibility of the NCC unit.

SYSTEM ANALYSIS

2.1 Existing System

The existing system of managing cadet activities in the Nirmala NCC unit is primarily manual, relying on paper-based records and registers. Cadet registrations are maintained in files, attendance is marked in physical sheets, and uniform bills are written by hand. While this traditional method is simple and requires no technical infrastructure, it poses several challenges such as time consumption, difficulty in retrieving past records, duplication of data, and a higher chance of human errors. Additionally, the lack of a centralized database makes it hard for administrators to manage and track information efficiently, especially as the number of cadets increases each year.

2.2 Proposed System

The proposed Nirmala NCC management system introduces a digital platform that replaces the traditional manual methods with a centralized, web-based solution. It allows cadets to register online, with admin approval to ensure authenticity, and provides a secure dashboard for both cadets and administrators. Attendance can be marked and tracked digitally, uniform items can be managed through an automated billing system, and all cadet details are stored in a structured database for easy retrieval. This system minimizes errors, reduces paperwork, saves time, and enhances transparency in all operations. By ensuring data security and accessibility from anywhere, the proposed system offers a more efficient, reliable, and modern approach to managing the activities of the Nirmala NCC unit.

2.3 System Requirement Specification (SRS)

2.3.1 Hardware Specification

- Processor: Intel i3 or above
- RAM: 4GB minimum
- Storage: 500GB HDD or SSD
- Monitor: 1024x768 resolution or higher

2.3.2 Software Specification

- Operating System: Windows/Linux
- Server: Apache (XAMPP/WAMP)
- Database: MySQL
- Language: PHP, HTML, CSS, JavaScript

2.3.3 Front End

HTML, CSS, JavaScript (with Bootstrap and Tailwind for styling).

2.3.4 Back End

PHP for server-side scripting and MySQL as the relational database.

2.4 Feasibility Analysis

- Technical Feasibility: Uses widely available technologies like PHP and MySQL.
- Operational Feasibility: Easy to operate for both admins and cadets.
- Economic Feasibility: Minimal cost since open-source technologies are used.

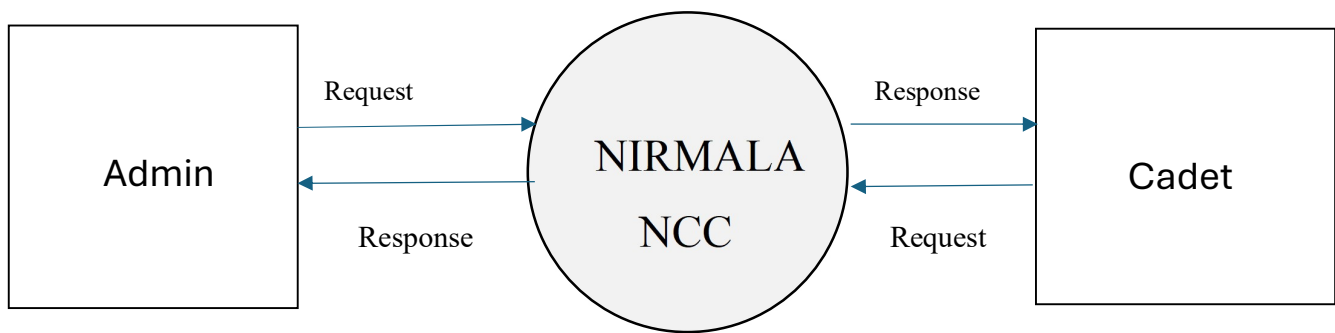
2.5 Data Flow Diagram

2.5.1 Introduction to Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation used to depict how data moves through a system, showing the flow of information between different processes, data stores, and external entities. It is a widely used tool in system analysis and design, as it provides a clear and structured way to visualize how inputs are transformed into outputs through various processes. A DFD helps both technical developers and non-technical stakeholders understand the overall structure and functioning of the system without going into programming details.

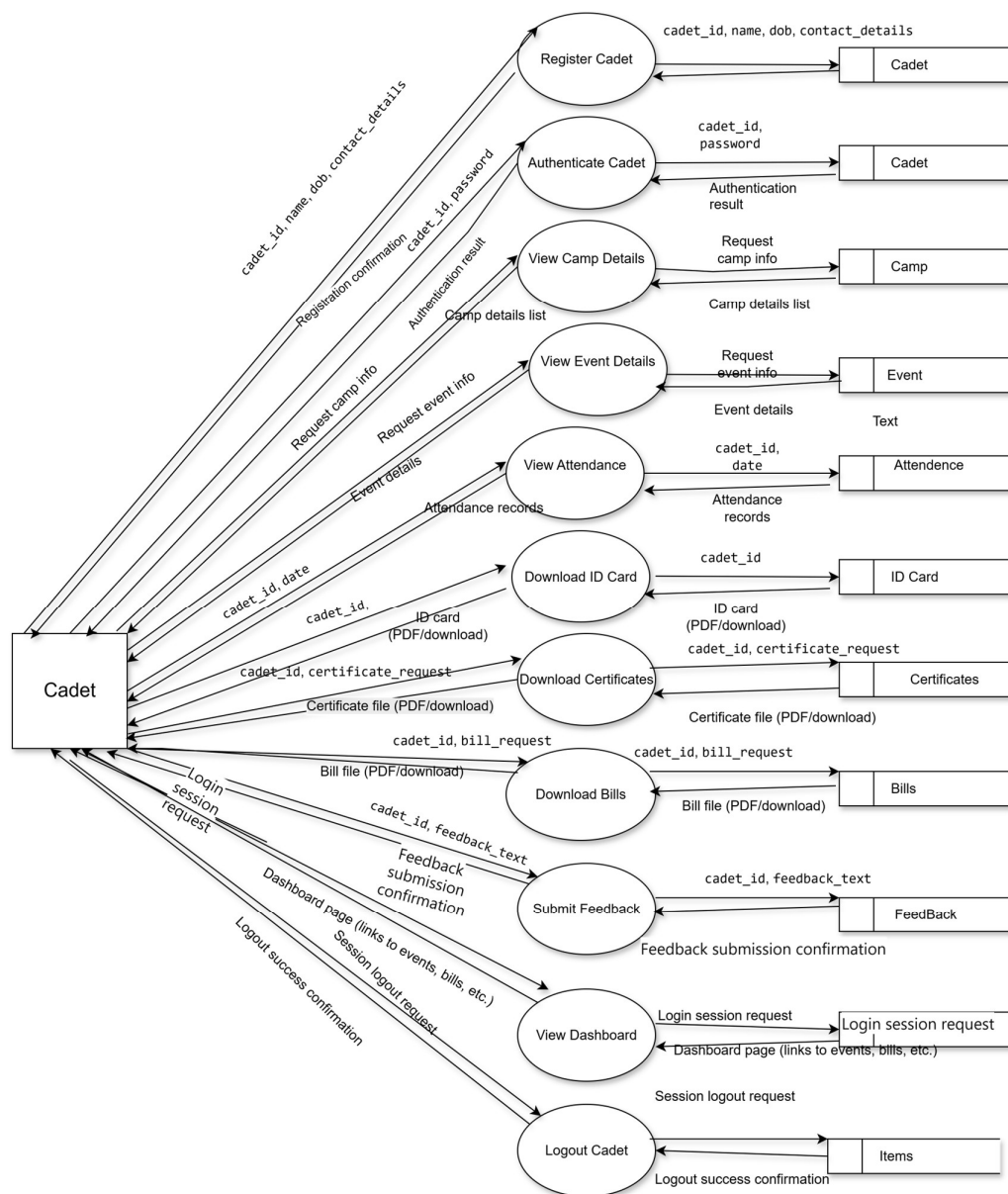
DFDs are generally created in different levels of detail, starting with the **Context Level Diagram (Level 0)**, which gives a broad overview of the entire system and its interaction with external entities. This is further expanded into **Level 1 and Level 2 diagrams**, which break down the processes into more detailed sub-processes, data flows, and storage points. By using these levels, a DFD provides both a high-level and detailed view of the system's functionality.

The main elements of a DFD include **processes** (represented by circles or rounded rectangles), **data flows** (arrows showing the movement of data), **data stores** (open-ended rectangles showing where data is stored), and **external entities** (squares representing sources or destinations outside the system). Together, these components make the DFD a powerful tool for analyzing existing systems, identifying inefficiencies, and designing effective new solutions.

LEVEL 0

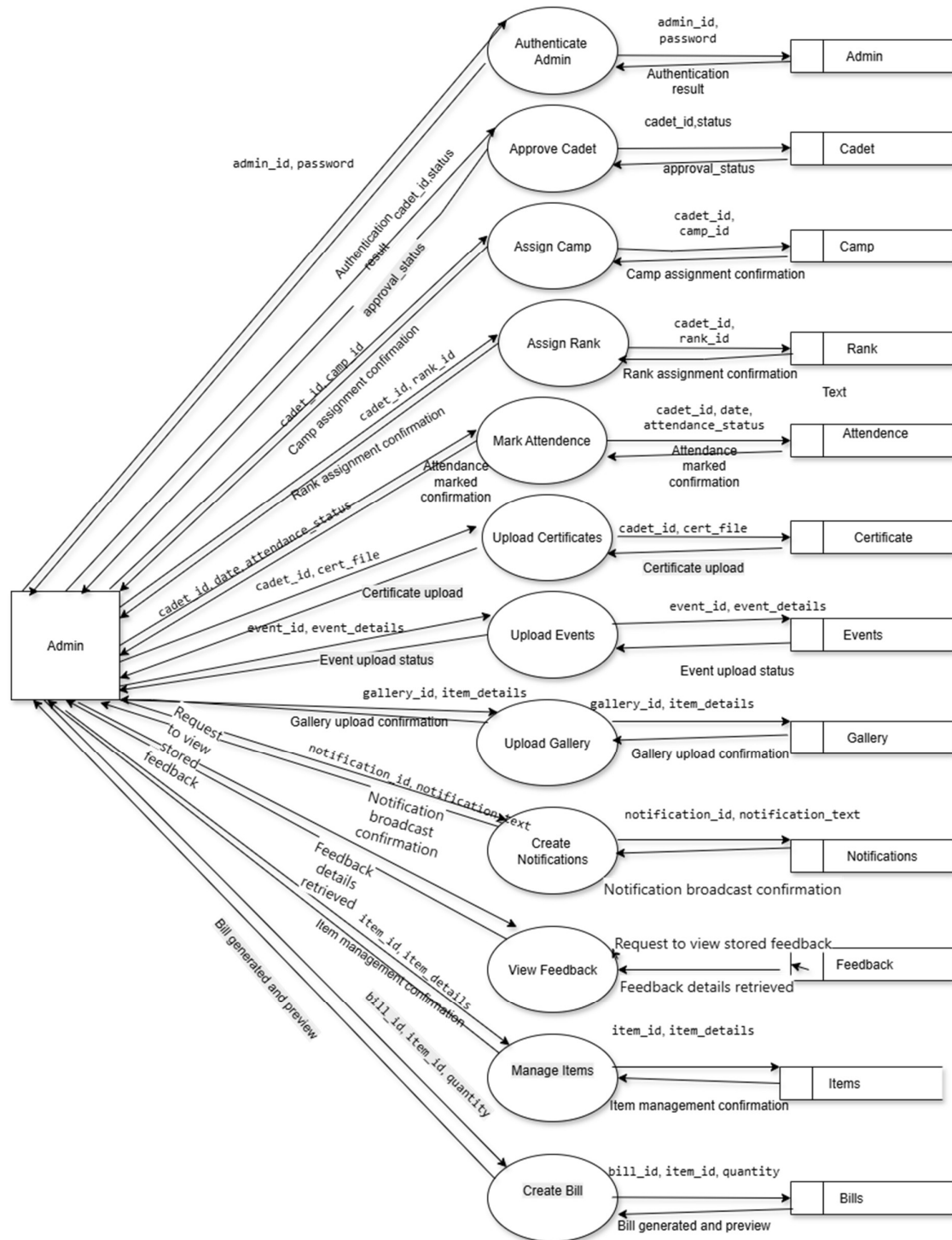
LEVEL 1

CADET



LEVEL 1

ADMIN



SYSTEM DESIGN

3.1 Input Design

The system provides forms for cadet registration, login, and attendance marking.

3.2 Output Design

The system generates dashboards, attendance reports, and ID cards.

3.3 Table Design

The database contains tables such as register, cadet details, attendance, and notifications.

admin

Column name	Data Type	Attributes	Description
admin_id	INT	Primary Key	Unique ID for each admin
username	VARCHAR(100)	NOT NULL	Admin username
password	VARCHAR(255)	NOT NULL	Hashed password
Created_at	TIMESTAMP	DEFAULT	When the record was created

a

user

Column name	Data Type	Attributes	Description
Cadet_id	INT	Primary Key	Auto Increment
Cadet_no	VARCHAR(50)	NOT NULL	UNIQUE
name	VARCHAR(100)	NOT NULL	Cadet name
email	VARCHAR(100)	NOT NULL	UNIQUE
phone	VARCHAR(10)	NOT NULL	UNIQUE
password	VARCHAR(255)	NOT NULL	Hashed password
status	ENUM('pending', 'approved', 'rejected')	Default	
role	ENUM('cadet', 'admin')	Default	
created_at	TIMESTAMP	Default	When the record was created

rank

Column name	Data Type	Attributes	Description
rank_id	INT	Primary Key	Unique ID for each admin
Cadet_no	VARCHAR(50)	NOT NULL, FOREIGN KEY	Cadet number linked to details
name	VARCHAR(150)	NOT NULL	Cadet name
rank	VARCHAR(100)	NOT NULL	Ncc rank
photo	VARCHAR(255)	NOT NULL	File path of uploaded cadet photo
year	VARCHAR(20)	NOT NULL	Year/batch
Created_at	TIMESTAMP	DEFAULT	When the record was created

Camps

Column name	Data Type	Attributes	Description
camp_id	INT	Primary Key	Unique ID for camp
Camp_name	VARCHAR(200)	NOT NULL	Name of the camp
location	VARCHAR(200)	NOT NULL	Camp location
Start_date	DATE	NOT NULL	Start date
End_date	DATE	NOT NULL	End date
description	TEXT	NULL	Camp description
camp_pdf	VARCHAR(255)	NULL	Uploaded camp PDF path
created_at	TIMESTAMP	Default	When the record was created

Cadet_camps

Column name	Data Type	Attributes	Description
id	INT	Primary Key	Unique ID
Cadet_no	VARCHAR(50)	NOT NULL, FOREIGN KEY	Assigned cadet
Camp_id	INT	NOT NULL, FK	Assigned camp
status	ENUM('upcoming', 'completed')	Default	Assigned status
assigned_at	TIMESTAMP	Default	Assignment date

Details

Column name	Data Type	Attributes	Description
cadet_no	VARCHAR(50)	Primary Key	Unique cadet number
fcadet_no	VARCHAR(100)	NULL	Full cadet number
Name	VARCHAR(150)	NOT NULL	Cadet name
Phone	VARCHAR(20)	NULL	Cadet mobile number

Dob	DATE	NULL	Date of birth
Batch	VARCHAR(20)	NULL	Batch info
Photo	VARCHAR(255)	NULL	File name of uploaded photo
Address	TEXT	NULL	Cadet address
Blood group	VARCHAR(10)	NULL	Blood group
College_ reg_no	VARCHAR(50)	NULL	College registration number
Class_ roll_no	VARCHAR(50)	NULL	Class roll number
Course	VARCHAR(100)	NULL	Course/stream
NCC_rank	VARCHAR(50)	NULL	NCC rank
Date_of_enrolment	DATE	NULL	Enrollment date
gender	EBUM('female', 'male', 'other')	NULL	gender
Identification_mark	VARCHAR(255)	NULL	ID mark
Height	INT	NULL	Height(cm)
Weight	INT	NULL	Weight(kg)
Parent_name	VARCHAR(150)	NULL	Parent's name
Parent_mobile	VARCHAR(20)	NULL	Parent's mob number

Attendance

Column name	Data Type	Attributes	Description
Attendance_id	INT	Primary Key	Unique
Cadet_no	VARCHAR(50)	NOT NULL,FK	Cadet number
Date	DATE	NOT NULL	Attendance date
Status	ENUM('present', 'absent')	NOT NULL	Attendance status
Marked_at	TIMESTAMP	Default	Time attendance was recorded

Add Item

Column name	Data Type	Attributes	Description
Item_id	INT	PK	Unique
Item_name	VARCHAR(150)	NOT NULL	Name of uniform item
Price	DECIMAL(10.2)	NOT NULL	Price of the item
Stock	INT	Default	quantity available in stock
Created_by	INT	FK	Admin who added the item
Created_at	TIMESTAMP	default	When the item was added

Uniform_items

Column name	Data Type	Attributes	Description
Item_id	INT	PK	Unique
Item_name	VARCHAR(150)	NOT NULL	Name of uniform item
Price	DECIMAL(10.2)	NOT NULL	Price per unit
stock	INT	default	Quantity in stock
Created_at	TIMESTAMP	default	Timestamp of creation

Uniform bills

Column name	Data Type	Attributes	Description
Bill_id	INT	PK	unique
Cadet_id	INT	NOT NULL,FK	Cadet for whom the bill is made
Total_amount	DECIMAL(10.2)	NOT NULL	Total amount of the bill
Bill_date	DATE	Default	Date of bill creation
Paid_status	ENUM('paid', 'unpaid')	Default	Payment status
Created_at	TIMESTAMP	default	Timestamp of bill creation

Certificates

Column name	Data Type	Attributes	Description
<u>cert_id</u>	INT	PK	unique
<u>Cadet_no</u>	<u>VARCHAR(50)</u>	<u>NOT NULL,FK</u>	
<u>Cert_name</u>	<u>VARCHAR(100)</u>	<u>NOT NULL</u>	
<u>Cert_type</u>	<u>VARCHAR(50)</u>	<u>NOT NULL</u>	<u>Used tp categorized</u>
<u>Cert_file</u>	<u>VARCHAR(255)</u>	<u>NOT NULL</u>	<u>certificates</u>
<u>Upload_date</u>	<u>DATETIME</u>	<u>default</u>	

Event

<u>Column Name</u>	<u>Data Type</u>	<u>Constraints / Notes</u>
<u>camp_id</u>	<u>INT</u>	<u>Primary Key, Auto Increment</u>
<u>camp_name</u>	<u>VARCHAR(200)</u>	<u>NOT NULL, name of the camp</u>
<u>location</u>	<u>VARCHAR(200)</u>	<u>NOT NULL, camp location</u>
<u>start_date</u>	<u>DATE</u>	<u>NOT NULL, start date of the camp</u>
<u>end_date</u>	<u>DATE</u>	<u>NOT NULL, end date of the camp</u>
<u>description</u>	<u>TEXT</u>	<u>Optional, description of the camp</u>
<u>camp_pdf</u>	<u>VARCHAR(255)</u>	<u>Optional, file path to uploaded camp PDF</u>
<u>created_at</u>	<u>TIMESTAMP</u>	<u>Default CURRENT_TIMESTAMP, record creation time</u>

Gallery

<u>Column Name</u>	<u>Data Type</u>	<u>Constraints / Notes</u>
<u>camp_id</u>	<u>INT</u>	<u>Primary Key, Auto Increment</u>
<u>camp_name</u>	<u>VARCHAR(200)</u>	<u>NOT NULL, name of the camp</u>

<u>location</u>	<u>VARCHAR(200)</u>	<u>NOT NULL, camp location</u>
<u>start_date</u>	<u>DATE</u>	<u>NOT NULL, start date of the camp</u>
<u>end_date</u>	<u>DATE</u>	<u>NOT NULL, end date of the camp</u>
<u>description</u>	<u>TEXT</u>	<u>Optional, description of the camp</u>
<u>camp_pdf</u>	<u>VARCHAR(255)</u>	<u>Optional, file path to uploaded camp PDF</u>
<u>created_at</u>	<u>TIMESTAMP</u>	<u>Default CURRENT_TIMESTAMP, record creation time</u>

Notification

<u>Column Name</u>	<u>Data Type</u>	<u>Constraints / Notes</u>
<u>camp_id</u>	<u>INT</u>	<u>Primary Key, Auto Increment</u>
<u>camp_name</u>	<u>VARCHAR(200)</u>	<u>NOT NULL, name of the camp</u>
<u>location</u>	<u>VARCHAR(200)</u>	<u>NOT NULL, camp location</u>
<u>start_date</u>	<u>DATE</u>	<u>NOT NULL, start date of the camp</u>
<u>end_date</u>	<u>DATE</u>	<u>NOT NULL, end date of the camp</u>
<u>description</u>	<u>TEXT</u>	<u>Optional, description of the camp</u>
<u>camp_pdf</u>	<u>VARCHAR(255)</u>	<u>Optio_</u>

4. SYSTEM TESTING & IMPLEMENTATION

4.1 System Testing

The system undergoes unit testing, integration testing, and user acceptance testing.

4.2 System Implementation

The project is deployed using Apache server with MySQL database on a local/remote environment.

5. SECURITY TECHNOLOGIES & POLICIES

The system ensures secure login with password hashing, session management, and role-based access control.

MAINTENANCE

The system can be updated with new features, bug fixes, and database optimizations.

SCOPE FOR FUTURE ENHANCEMENTS

Mobile application, SMS notifications, and integration with college ERP can be added.

CONCLUSION

The Nirmala NCC project successfully digitizes cadet management, improves efficiency, and reduces manual workload.

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APPENDIX

10.1 Screenshots of the system interfaces

Login

Registration

Dashboard

Attendance

ID Card